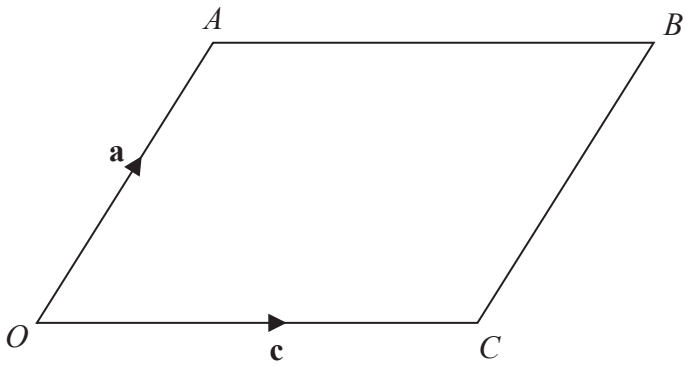
**19**




$OABC$ is a parallelogram.

$\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$

$X$ is the midpoint of the line $AC$.
$OCD$ is a straight line so that $OC : CD = k : 1$

Given that $\overrightarrow{XD} = 3\mathbf{c} - \frac{1}{2}\mathbf{a}$

find the value of $k$.

Bold letters represent vectors.

OX = ½a + ½c  (X is the midpoint of AC)

**1 mark**
Use the midpoint to form a vector

OD = OX + XD

= ½a + ½c + 3c – ½a = 3.5c

CD = OD – OC = 3.5c – c = 2.5c

**1 mark**
Find CD correctly

OC : CD = 1 : 2.5 = k : 1

**1 mark**
Use the stated ratio to find k

Divide both parts by 2.5:

k = 1 ÷ 2.5 = $\frac{2}{5}$

**1 mark**
Correct value

$k =$ $\frac{2}{5}$

**(Total for Question 19 is 4 marks)**

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